

New 3D Chipsets on the Horizon				
Manufacturer	nVidia	ATi	Matrox	Creative/3Dlabs
Chip name	NV30	R300	Parhelia-512	P10
Cards shipping	Q4 2002	Q3 2002	Q3/Q4 2002	Q3/Q4 2002
Manufacturing process	0.13 micron	0.15 micron	0.15 micron	0.15 micron
DirectX support	DirectX 9	DirectX 9	DirectX 8.1, and preliminary DirectX 9	DirectX 8.1, and preliminary DirectX 9
Output support	Multihead, DVI/TV-Out	Multihead [HydraVision] DVI/TV-Out	Multihead [TripleHead & Surround Gaming], DVI/TV-Out	Unknown
Memory support	256-bit DDR 256 MB	128-bit DDR [possibly 256-bit DDR]	256-bit DDR 256 MB	256-bit DDR 256 MB
Rendering pipelines	8	8	4	4
Pixel shaders	2 x DirectX 9	4 x DirectX 9	4 x DirectX 8 (PS 1.3)	4 x DirectX 8 (PS 1.3)
Vertex shaders	4 x DirectX 9	2 x DirectX 9	4 x DirectX 9	4 x DirectX 9
AGP support	AGP8X	AGP8X	AGP8X	AGP4X
Antialiasing	8x AA	ATi Smoothvision Supersampling	Edge AA (16x)	Programmable Edge AA, 8x Multisampling
Rendering precision	64-bit (floating point)	64-bit (floating point)	40-bit (fixed)	64-bit programmable (fixed)

to effectively act as multiple virtual VPUs, a technique very similar to Intel’s hyperthreading for Pentium 4 CPUs. It allows for fast task-switching between multiple 3D applications and plays an active role in any multithreaded 3D application where the VPU has to work upon multiple threads. This will be used extensively in Microsoft’s next generation operating system, Longhorn, but doesn’t make a great deal of sense from today’s standpoint.

Wait, there’s more

The P10 interfaces with its DDR memory using a 256-bit memory controller interface, as opposed to the current generation 128-bit architecture. This gives the P10 a memory bandwidth of 20 GBps. Moreover, 3Dlabs offers a Virtual Memory Architecture that is quite similar in technique to how current CPUs manage mem-

ory. It allows the chip to seamlessly access textures regardless of whether they are physically present in the video card memory (locally) or in the system RAM. As you can see, the P10 really goes a long way in blurring the line between a CPU and a graphics chip.

API support

The P10 supports DirectX 8 and its support of DirectX 9 extends to the new vertex shader implementation (Vertex Shader 2.0), but stops short of Pixel Shader 2.0 support. Since 3Dlabs has taken an initiative in defining the OpenGL 2.0 standard, it is to be expected that the chip will support the standard to the fullest extent possible. Furthermore, the flexibility granted to the P10 by its programmable architecture, should ensure that any future additions to either DirectX 9.0 or OpenGL 2.0 could be easily implemented into the chip.

What lies ahead

The game industry is very unkind to newcomers. Unless the product has a significant market share, no developer is going to spend time and money developing their next big thing on new hardware. This is why something seemingly trivial as 32-bit colour in 3D games took as long as it did to arrive. This is also why it is only now that we see

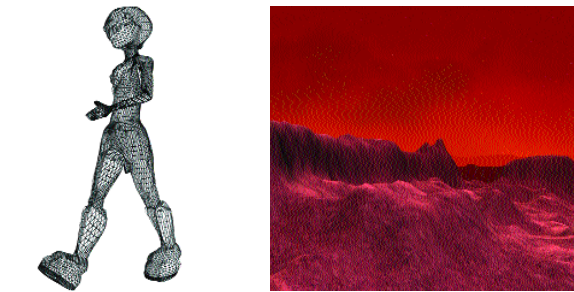
titles demanding T&L hardware and 32 MB of onboard RAM. What all of this implies is that a newcomer will have to toe the API line—any innovative feature introduced in their chip will be stillborn unless it is supported by at least one, if not both DirectX 9 and OpenGL 2.0 APIs. And this will possibly be the biggest challenge to counter.

No sleeping giants, these

Meanwhile the leaders, nVidia and ATi, are busy trying to best each other. nVidia has fallen behind its six-month product cycle, some say no thanks to the Xbox; and ATi has taken every advantage of this fact. ATi’s latest offering, codenamed the R300 is slated to be released in a few months, while nVidia’s next chip, the NV30, isn’t expected to go into production until late this year. So if everything goes right for ATi, they have nothing but a possibly faster GeForce 4 part from nVidia, to come between them and the pole position—at least for a period of four to five months, by which time the NV30 should be out.

It will have to be amidst this war that the new contenders would have to unfurl their own banners, and hope that the market pays attention. It is going to be a long and bloody war, fought on fronts ranging from technology, to driver quality and game compatibility, to the all-important price factor. It might get messy, but the ultimate winner will be the consumer. 

HATIM KANTAWALLA and AHMED SHAIKH



The pictures above show how Parhelia handles Hardware Displacement Mapping (HDM) by creating a mesh and superimposing a displacement map on it. Although the P10 doesn’t count HDM as a unique feature, the 3Dlabs chip can produce similar results using its programmable architecture